ROBOCORE ZAMBIA

RoboCore Zambia

PLASTAL-BOT BUILDERS

GET STARTED HERE

www.plastalbotbuilders.com

On this page you will find:

- **Assembly and first-run guide**
- **Step-by-step classroom lessons**
- **Programming environments: blocks, Python, C++**
- **Firmware, example code and libraries**
- **3D-print files for the chassis and the arm**
- **Support forum and teacher resources**

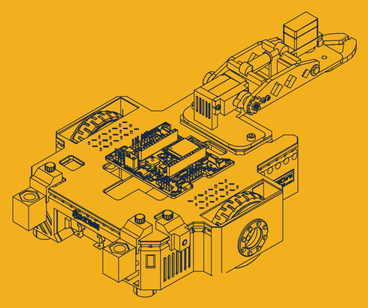
BEFORE YOU START

- 3 x 18650 lithium cells (not included) and a USB-C charger
- A computer or tablet with a browser, or the Arduino IDE
- Charge the robot fully before the first lesson
- Fit the arm module only after the drive base runs

Replacement parts, add-on sensors and the arm module:
plastalbotbuilders.com/robocore

Design files are published. Print it, fix it, improve it, share it.

SCAN FOR DOCS, CODE AND LESSONS



PLASTAL-BOT BUILDERS

Plastal-Bot Builders is a Lusaka makerspace that teaches robotics, coding and fabrication to learners and teachers across Zambia. RoboCore is our own controller family, designed so a class can build, program and repair its robots locally.

WORKSHOPS · TEACHER TRAINING · ROBOTICS CLUBS · FABRICATION

www.plastalbotbuilders.com

SCAN

JOIN THE ROBOCORE COMMUNITY

Share what your class builds, find lesson plans and compare notes with other schools. Replacement parts, add-on sensors and the arm module are available from us; the design files are published so you can also print and fix parts yourself.

LESSON PLANS · FORUM · SPARE PARTS · DESIGN FILES

www.plastalbotbuilders.com/community

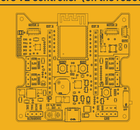
SCAN

THIS KIT INCLUDES

RoboCore V2 robot, assembled



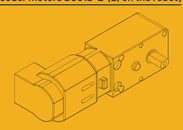
RoboCore V2 controller (on the robot)



3D-printed chassis (on the robot)



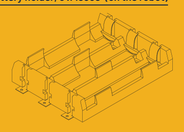
Encoder motors DG01D-E (2, on the robot)



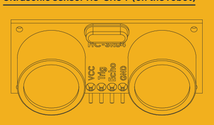
Wheels 56 mm, TPU tyres (2)



Battery holder, 3 x 18650 (on the robot)



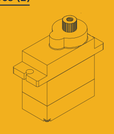
Ultrasonic sensor HC-SR04 (on the robot)



Line sensor, 3 channels (on the robot)



Micro servos (2)



Arm boom



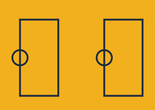
Arm base



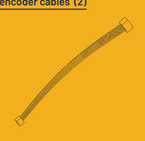
Claw jaws (2)



Ballast clips (2)



Motor encoder cables (2)



USB-C cable



Screws, nuts and quick-start card



For replacement parts and the arm module visit plastalbotbuilders.com/robocore

Designed and built in Zambia by Plastal-Bot Builders, Lusaka

ROBOCORE V2 · BUILD. CODE. CREATE.

INSIDE (mirrored: prints on the back of the outside sheet). RoboCore V2 mailer box, roll-end tuck-top, outer 273 x 203 x 83 mm, E-flute 1.5 mm. Print 1:1; navy solid = cut, grey dashed = fold. Bar = 100 mm.